CubeSat-BASED 5G CLOUD RADIO ACCESS NETWORKS

A Novel Paradigm for On-Demand Anytime/Anywhere Connectivity

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5G networks are expected to become the main infrastructure for security verticals such as disaster relief, humanitarian aid, and governmental and defense communications. In the case of defense services, especially, there are complex areas where coverage and connectivity are not reliable or are completely absent. Thus, the deployment of drones as mobile base stations (BSs) also needs a system designed for reliable backhaul based on satellites. However, a baseband unit (BBU) [devoted to baseband signal processing at the radio access network (RAN)] on drones is

Digital Object Identifier 10.1109/MVT.2020.2979056
Date of current version: 25 March 2020

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A CubeSat is a kind of standardized satellite that belongs to the family of small satellites (i.e., satellites that weight under 300 kg).

not a flexible solution and may require great power supply and processing capabilities, which a drone can hardly host.

The work we discuss here studies and designs a cloud RAN system based on mobile BSs and CubeSats, where BBUs are virtualized on low-cost satellites. In particular, the work takes advantage of a split of virtual BBUs to virtualize upper layers on CubeSats, while guaranteeing the satisfaction of technical requirements.

Background

The main aim of future 5G and beyond (B5G) networks is to create an ecosystem capable of concurrently supporting heterogeneous services with different requirements. Aside from civilian services such as media distribution, smart transportation, smart cities, and smart industry, 5G and B5G networks are also expected to host services related to governmental and defense activities, emergency and safety, disaster relief, and humanitarian aid. Within this context, border monitoring and security represent activities that will be included in such future networks' ecosystem.

Nevertheless, hosting such critical verticals requires warranting 100% coverage and network availability to ensure anytime/anywhere connectivity [1]; this implies the effective and efficient provisioning of network resources for reliable communications not only in urban and populated areas but also in rural and complex areas, i.e., where existing terrestrial networks are absent or cannot properly work. For example, a specific use case can be one of complex borders, where monitoring technologies cannot rely on or connect to legacy terrestrial wireless networks. In such contexts, current solutions suggest the deployment of either direct satellite connection to terrestrial end users or mobile BSs (drones).

Recently, the deployment of unmanned aerial vehicles (UAVs) has become an efficient solution to provide on-demand network coverage and additional security, with drones used as monitoring aerial peripherals and/or mobile BSs for connectivity provisioning. However, a significant challenge can arise, such as whether to run baseband processing when UAVs are used in areas without reliable (or any) connectivity (i.e., complex borders).

When providing cellular-like coverage in areas such as complex borders via UAVs, the legacy solution has been to implement a full BS, but an issue can arise because baseband processing is the most complex part of the RAN. The BBU is the entity responsible for performing baseband processing. Both a BBU and the remote radio head (RRH) constitute the cellular BS. Adding a BBU to each drone does not represent an efficient solution. Side by side, the UAVs should also support satellite backhaul connections to allow for communications between mobile BSs and satellite-based core networks. Nevertheless, when mobile BSs are deployed, virtualizing BBUs can be useful to move processing and load to centralized entities, which have higher capabilities but do not have the same battery constraints of drones.

Given the aforementioned considerations, the solution of virtualizing BBUs on satellites may emerge. Nevertheless, baseband processing has very strict and stringent requirements; as a result, its implementation on satellite-based virtual BBUs is not straightforward.

A preliminary structure of the proposed system was first presented in [2]; however, that work considers drones equipped with LoRaWAN gateways, without providing a detailed system-level description and evaluations. In fact, the scope of [2] is the estimation of an upper bound on the power gain provided by the virtualization of BBUs. Moreover, the analysis in [2] neglects the problem of a virtual BBU split with respective requirements in terms of latency and throughput.

The current article proposes and presents a detailed system-level analysis of a novel satellite-based approach to cloud RAN (C-RAN). Here, we describe the characteristics of a UAV-CubeSat architecture to design and perform lower-layer baseband functions at the drones while moving upper-layer baseband functions to satellites. The proposed design allows for CubeSats and UAVs that provide effective connectivity to terrestrial end users or peripherals, guaranteeing the satisfaction of C-RAN requirements on the fronthaul/backhaul in terms of latency and throughput. In particular, the article deals with system-level design and evaluation, focusing on a UAV-CubeSat system to support specification, which is needed for the virtualization of BBUs.

System Preliminaries

The proposed C-RAN system is mainly composed of three layers: terrestrial, aerial, and satellite based. The satellite-based layer includes small satellites that provide computing resources to allow for the virtualization of RAN functions. CubeSats are used at the third layer because these satellites have recently become very popular due to their low deployment cost and high flexibility.

A CubeSat [3] is a kind of standardized satellite that belongs to the family of small satellites (i.e., satellites that weight under 300 kg). A small satellite has to satisfy specific requirements in terms of shape, size, and weight. A single-unit CubeSat has a cubic shape

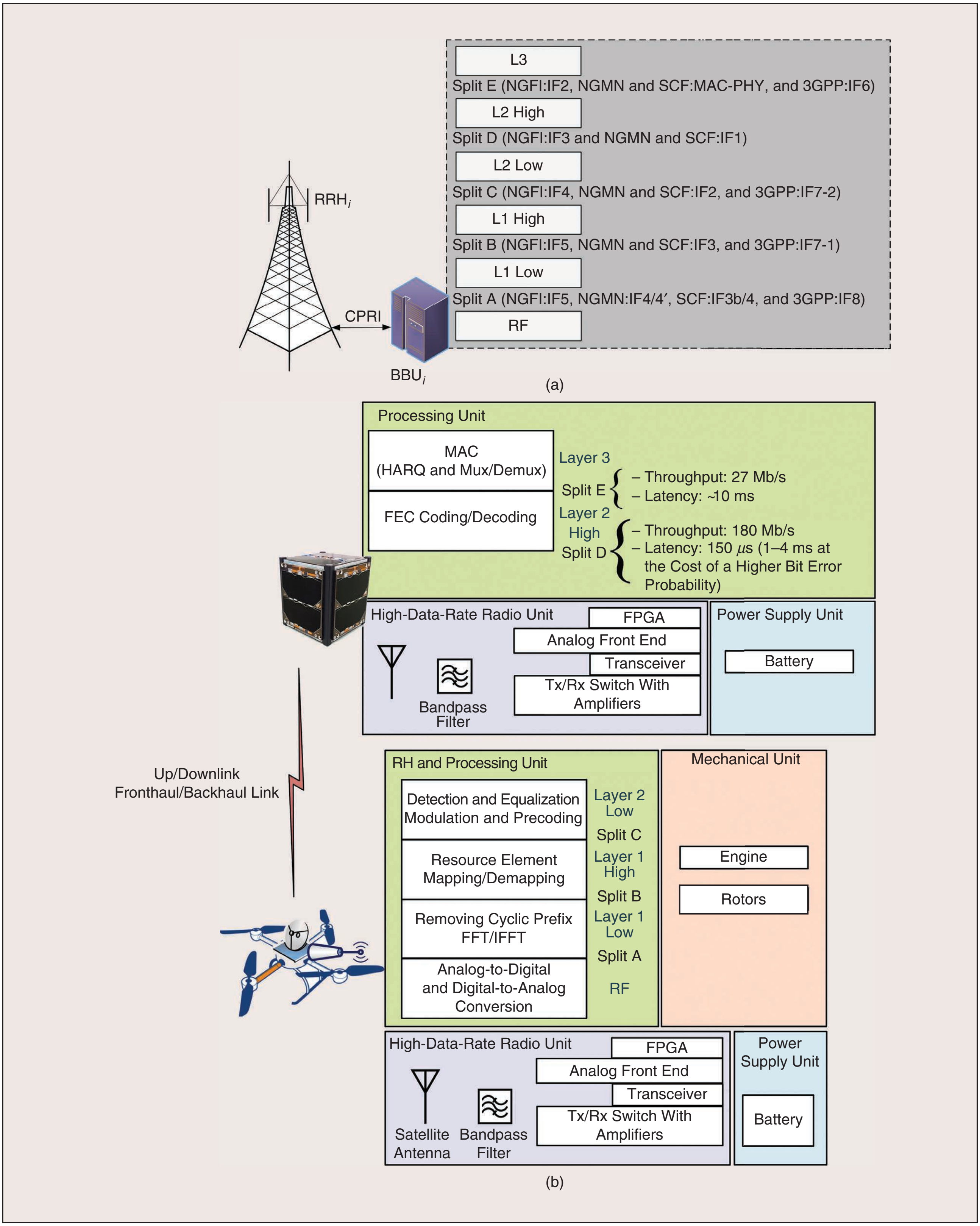

FIGURE 1 (a) The structure of a legacy 4G RAN with the representation of a BBU logic split. Different notations are included for completeness. (b) A diagram representing the detailed architecture of the proposed C-RAN system and its main characteristics. NGFI: Next Generation Fronthaul Interface; NGMN: Next-Generation Mobile Network; SCF: small cell forum; MAC: medium access control; PHY: physical layer; 3GPP: 3rd Generation Partnership Project; Mux/Demux; multiplexer/demultiplexer; RF: radio frequency; Tx: transmitter; Rx: receiver; FPGA: field-programmable gate array.

REGARDING THE COMMUNICATION BETWEEN UAVS AND CUBESATS, IT IS IMPORTANT TO NOTE THAT BOTH DEVICES MOVE, ONE WITH RESPECT TO THE OTHER ONE.

measuring $10\,\text{cm} \times 10\,\text{cm} \times 11\,\text{cm}$, with a mass of approximately 1–1.33 kg. Then, from that basic unit, larger standard CubeSats have been proposed, such as ones sized for 1.5, 2, 3, and 6 U.

As mentioned previously, in current 4G cellular networks, the BBU is the core processing entity of the RAN. Figure 1(a) displays the composition of a 4G RAN and legacy BBU. The RRH is located on the radio site and consists of transmitting/receiving antennas, while the BBU contains lower-layer processing and hybrid automatic repeat request (HARQ). Two components, the RRH and BBU, are connected to each other via a cable (normally fiber) supporting the Common Public Radio Interface (CPRI) standard. This standard seeks to guarantee a bit error rate (BER) for a data and control plane lower than 10^{-12}, and it employs a forward error correction (FEC) based on Reed–Solomon (RS) codes [4].

In such a context, C-RAN was proposed to centralize baseband processing, while avoiding the presence of a physical BBU at each radio site. In this way, virtual BBUs can run on general-purpose servers at data centers in the operators' core network, which also implies the optimization of the assignment of resources, scalability, and energy efficiency. Furthermore, in the last decade, virtual BBU splitting has aroused significant interest [5], [6]. The concept of a BBU split starts from the idea that BBU functions can be divided into logic subfunctions that can be run as separate entities. However, interfunction communications have specific and very stringent requirements in terms of latency and throughput, depending on the specific split chosen.

Figure 1(a) also shows the legacy solution for a logic split of BBU internal functionalities [6], [7]. The lower layer consists of radio-frequency processing, used for the analog-to-digital and digital-to-analog conversion of received signals. Above that, there are five layers, each of which has its own specific requirements to guarantee seamless communications. Layer 1 low (L1 low) is responsible for the cyclic redundancy check, fast Fourier transform (FFT), and inverse FFT (IFFT). The separation of these layers is called *split A*. Its required latency must be 150 µs, with throughput equal to 2,457 Mb/s [6]. L1 high is devoted to the mapping/demapping of resource elements. Split B is the division between L1 low and L1 high, which requires throughput equal to 720 Mb/s and a latency of 150 µs (with a possible relaxation in the case of slow-moving users and opportunistic HARQ at the cost of degraded performance).

Next, layer 2 low (L2 low) performs the modulation/demodulation and equalization of signals, while L2 high is responsible for FEC procedures. The split between L1 high and L2 low is called *split C* and needs throughput equal to 360 Mb/s and a latency equal to 150 μs (with a possible relaxation, such as in the cases mentioned previously). On the other hand, split D separates L2 low and L2 high. Finally, layer 3 (L3) manages HARQ with the transmission/reception of acknowledgements/negative acknowledgements (ACKs/NACKs). Split E divides L2 high and L3. The requirements of splits D and E are represented in Figure 1(b).

Side by side and for completeness and clarity, Figure 1(a) also includes a notation of the various levels of split, as proposed by the Ethernet-based Next Generation Fronthaul Interface (Small Cell Forum), the Next Generation Mobile Network Alliance, and the Third Generation Partnership Project [8].

The whole baseband processing of legacy 4G networks requires a general threshold of 3 ms to be completed (it may be extended to 4 ms but at an eventual higher price in terms of errors) [6], [7]. Moreover, HARQ employs acknowledgments with FECs using concatenated convolutional codes of variable rates (normally between one third and one half) with turbo decoding. In particular, an L3 should target a BER lower than 10^{-5}. A possible alternative is to use HARQ with pure FEC coding without retransmissions. Such a solution can be of interest for long-range C-RAN settings as they avoid delays due to ACK/NACK transmissions (as previously demonstrated by [9] in the context of vertical handovers). In such a case, robust FEC coding with a low rate (one third or one half) should be adopted to guarantee powerful error correction capabilities and the avoidance of ARQ techniques.

Regarding the communication between UAVs and CubeSats, it is important to note that both devices move, one with respect to the other one. That implies the presence of a Doppler effect, which affects the frequency of the signal and should be taken into account during system design. Figure 2 shows the main elements required in the calculation of the Doppler effect (for simplicity, we consider the drone hovering) [10]. In particular, the Doppler shift, considering drones' communications with low-Earth-orbit satellites, becomes the product among the frequency of transmission (f), module of velocity $\vec{V}$, and cosine of θ angle, all divided by the speed of light (c).

The high-data-rate radio unit block in Figure 1 identifies that the hardware involved in the physical layer (PHY) is the satellite communication for the fronthaul/backhaul. The achievable data rate R [11] can be calculated as

$$R = rs^{-1}M, \tag{1}$$

where r is the symbol rate, s is the number of samples per symbol, and M is the number of modulation bits. Thus, a modulation with greater constellations can increase the data rate, but at the cost of a higher BER. In such a case, RS FEC coding, forecast by the CPRI standard, should be adequately designed. To have reliable, analogic front-end operations (e.g., analog-to-digital and digital-to-analog conversion), s is normally set to 3 [11]. Conversely, r is generally imposed by specific hardware constraints [11].

CubeSats for the Virtualization of RAN

As mentioned previously, the proposed C-RAN system for network access provisioning in complex areas is composed of three main layers. The terrestrial layer includes all end users consisting of things (e.g., cameras, sensors of different kinds, 3D optical scanners, monitoring buoys, and so on) or smart mobile devices (for human-to-human communications). All of these devices are connected via a wireless cellular link to the second layer. The aerial layer consists of mobile BSs, which are realized using the deployment of UAVs (i.e., multirotors or multicopters). These devices are equipped with two antennas: one for terrestrial RAN and the other for satellite backhaul as well as a processing unit for performing L1/L2 low procedures. The colored blocks of Figure 1 depict the logic parts of virtualized baseband processing and its 4G link, those referred to as the *satellite link*, and those devoted to power supply and mechanics. In this article, we focus on the virtualization of a BBU split D, thus automatically implying the support of split E.

Regarding UAVs, mechanical parts and power supply equipment affect not only the movement but also the performance and characteristics of communications. This occurs because engines and rotors consume battery power, and battery life is a key metric that guarantees reliable and acceptable transmissions. Finally, a satellite layer is composed of CubeSats. These satellites contain the radio antenna used for backhaul communications and the processing units hosting the virtual functions of L2 High and L3.

The fronthaul/backhaul link can suitably use 2,400–2,483.5 MHz of free frequency space in the S band of 2,300–2,450 MHz, which can be assigned to satellite applications (including CubeSats). In particular, the International Telecommunication Union recommends avoiding interference with terrestrial Wi-Fi and industrial, scientific, and medical services [12]; however, this concern does not affect the proposed system because it is focused on complex border monitoring, in areas without connectivity.

DOPPLER SHIFT CHANGES WITH TIME DUE TO THE VARIATION OF THE RADIAL SATELLITE VELOCITY INHERENT IN THE VARIATION OF THE ELEVATION ANGLE DURING ORBIT TRAVEL.

Next, the Doppler shift changes with time due to the variation of the radial satellite velocity inherent in the variation of the elevation angle during orbit travel. For this reason, the satellite receiver should be able to estimate and compensate for the Doppler variation, time after time, to guarantee the correct carrier synchronization.

System-Level Evaluation

In Figure 2, the importance of angle θ is represented from a Doppler perspective. As far as latency is concerned, given the characteristics of a CubeSat's orbit, the distance between the drone and satellite—and, therefore, the propagation delay—changes until reaching the minimum when the elevation angle α is 90°. Thus, the maximum delay is achieved when α is zero. To simulate a CubeSat's orbit and the characteristics of its trajectory, we take into account a circular orbit. The orbit simulation was done with MATLAB. In particular, Figure 3 shows the simulation of various satellites' orbits across complex areas in Europe, where the proposed system would be applicable to host a vertical

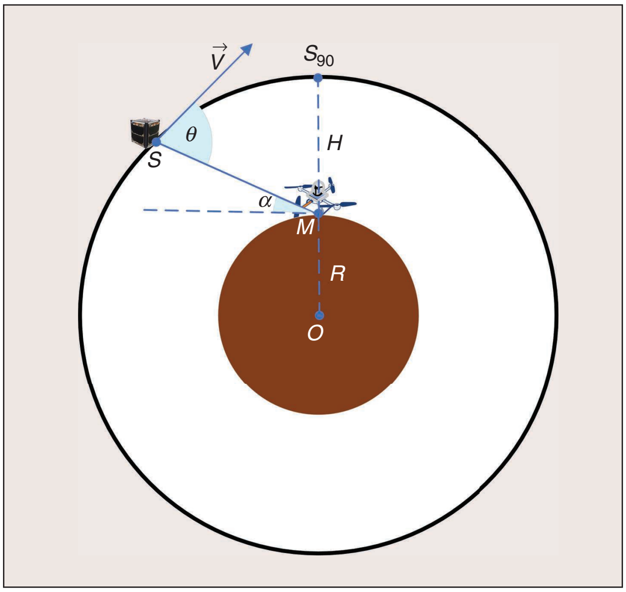

FIGURE 2 A simplified scheme of a CubeSat's relative movement with respect to a hovering UAV. The brown circle is the Earth, with center O and radius R. Point M is the position of the drone, while S is the position of the satellite (which changes in time). Point S_{90} is the position of satellite perpendicular to M.

THE SATELLITE RECEIVER SHOULD BE ABLE TO ESTIMATE AND COMPENSATE FOR THE DOPPLER VARIATION, TIME AFTER TIME, TO GUARANTEE THE CORRECT CARRIER SYNCHRONIZATION.

devoted to border-monitoring operations. Then, we consider three terrestrial points in three areas, which have to connect to the CubeSat. Furthermore, three curves were considered to guarantee a communication delay of between 1.5 and 3 ms, which satisfies the mandatory requirement for a latency lower than 4 ms for split D. Finally, given the defined ranges for the CubeSat's altitude, we reasonably consider an average distance between the CubeSat and Earth of between 250 and 600 km.

Figure 4 shows the variation of the maximum transmission angle according to the satellite's altitude (i.e.,

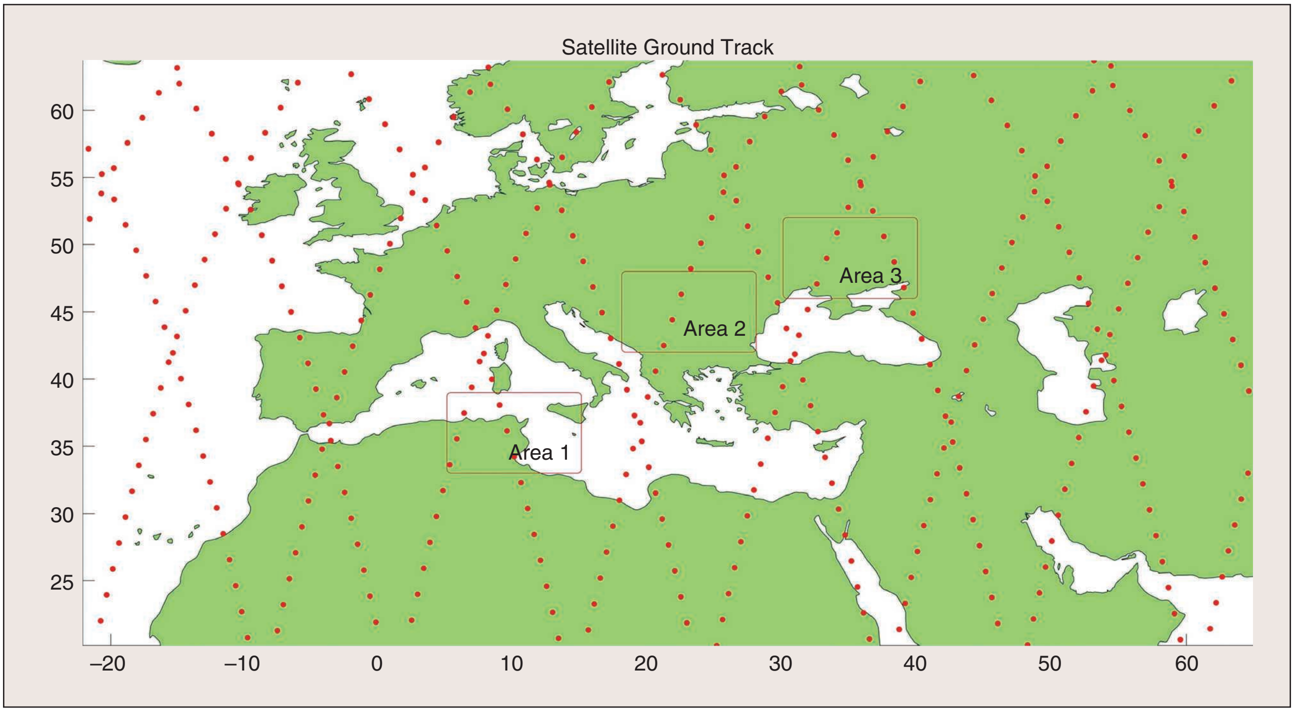

FIGURE 3 The scenario of simulation. The orbit of the CubeSat crosses the three main areas (1, 2, and 3), where there are complex European borders to be monitored.

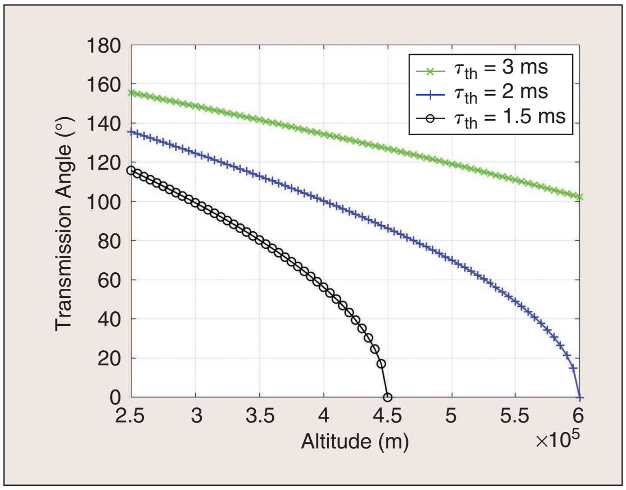

FIGURE 4 An evaluation of the maximum transmission angle between the UAV and CubeSat according to the altitude of the satellite's orbit. The curves are calculated for thresholds $\tau = 1.5$ ms, $\tau = 2$ ms, and $\tau = 3$ ms.

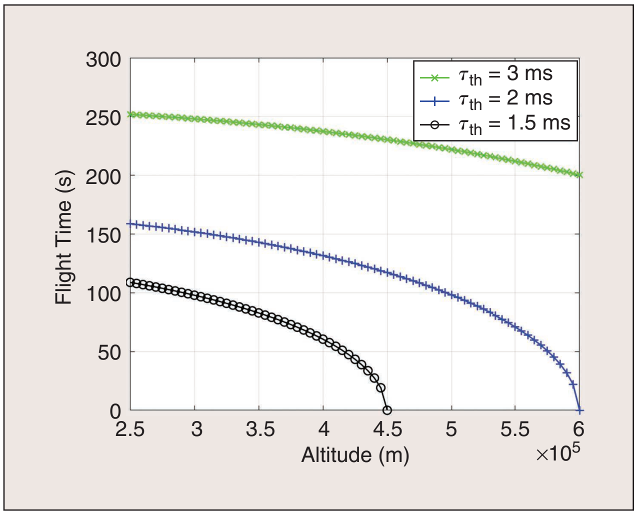

FIGURE 5 An evaluation of the maximum flight time of the CubeSat according to the altitude of the satellite's orbit. The curves are calculated for thresholds $\tau = 1.5$ ms, $\tau = 2$ ms, and $\tau = 3$ ms.

segment MS_{90}). The transmission angle is calculated as $2(90-\alpha)$, where α is the elevation angle, i.e., the angle between the two maximum acceptable distances to start and end the transmission because of propagation delay. It is possible to see that this angle almost linearly decreases if we allow for a delay of 3 ms, while it reduces more rapidly if we need to guarantee a lower threshold. Clearly, the use of lower orbits significantly increases the transmission angle. Indeed, given a fixed requirement on the maximum propagation delay (to allow the BBU to perform the virtualized task), the maximum acceptable distance from the drone to the CubeSat, namely, the slant range, is also fixed. This implies that, by lowering CubeSat's altitude, the elevation angle α must be reduced to obtain the required slant range. As a result, the transmission angle increases with the flight time accordingly.

In parallel, it is also important to look at the curves in Figure 5. By considering a reasonable average satellite speed of 7 km/s [10], the time for backhaul/fronthaul transmission diminishes from approximately 230 to 180 s (for $\tau_{th} = 3$ ms), while it falls from approximately 150 to 10–20 s (for $\tau_{th} = 2$ ms) and from 100 to 5–10 s (for $\tau_{th} = 1.5$ ms).

Next, Figure 6 focuses on the evaluation of the Doppler shift for $\tau_{th} = 3$ ms, $\tau_{th} = 2$ ms, and $\tau_{th} = 1.5$ ms. As expected, the Doppler shift inversely increases with the altitude and is in line with what was previously estimated in [10].

Finally, it is fundamental to analyze the impact of latency on the C-RAN's performance from the perspective of the LTE turbo-decoding operation performed by L2 High at the satellite. In [13], an initial study was presented for terrestrial, fiber-based virtual BBU–RRH connections. Turbo decoding is iterative, and the BER performance of the decoder essentially depends on the number of iterations that the remote BBU can execute within a maximum delay (namely, the delay budget) imposed by LTE network temporization constraints. For L2 High concerns, all of the decoding operations must be completed within a mandatory delay budget of 4 ms [6]. In this article, we consider the more reliable value of 3 ms, as discussed in [13].

To avoid the transmission/reception of ACKs/NACKs, the work presented here considers pure FEC for reliable L3 transmissions. Under these conditions, the number of turbo-decoding iterations that the remote BBU can support is given as [13]

CLEARLY, THE USE OF LOWER ORBITS SIGNIFICANTLY INCREASES THE TRANSMISSION ANGLE.

$$k = \max\left\{0, \left\lfloor \frac{pO(\Phi - J - D - T_{\text{proc}})}{LF} \right\rfloor\right\}, \tag{2}$$

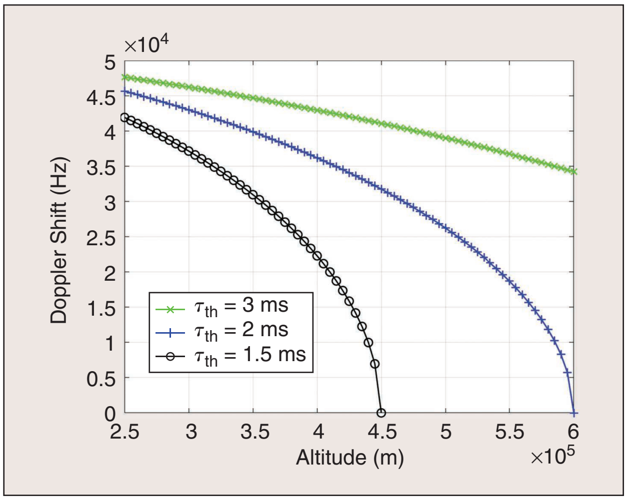

FIGURE 6 An evaluation of the maximum Doppler shift because of CubeSat's relative movement with respect to the UAV according to the altitude of the satellite's orbit. The curves are calculated for the thresholds $\tau = 1.5$ ms, $\tau = 2$ ms, and $\tau = 3$ ms.

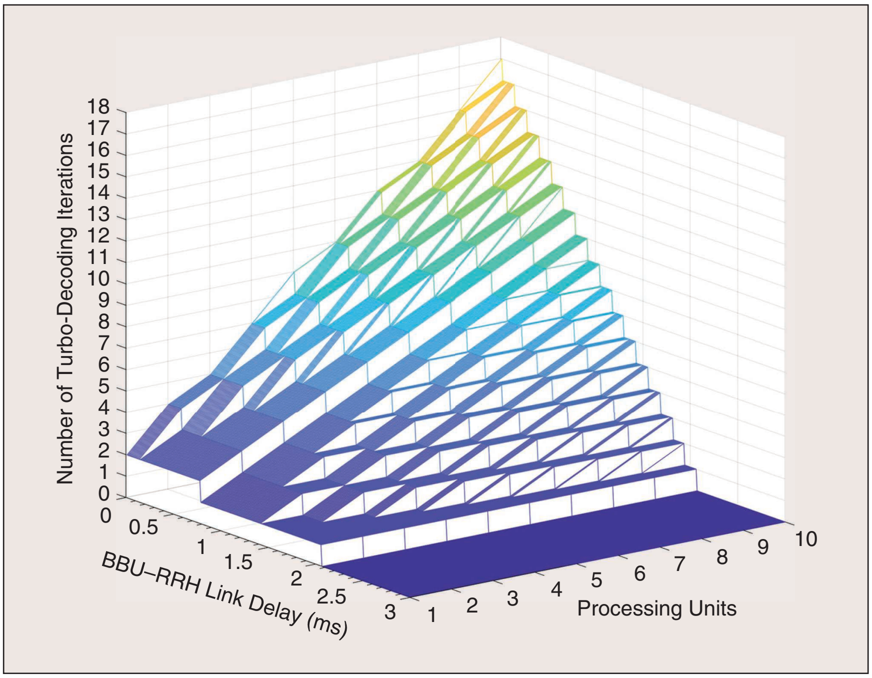

FIGURE 7 The number of turbo-decoding iterations allowed in the case of split D of the virtual BBU.

TO THE BEST OF OUR KNOWLEDGE, THE WORK DESCRIBED IN THIS ARTICLE IS THE FIRST THAT DEALS WITH THE DESIGN OF C-RAN AND A VIRTUAL BBU SPLIT USING CUBESATS.

where:

- p is the computational power of a single-core processor in giga operations per second (GOPS), O is number of core processors installed inside the BBU, and Φ is the delay budget.

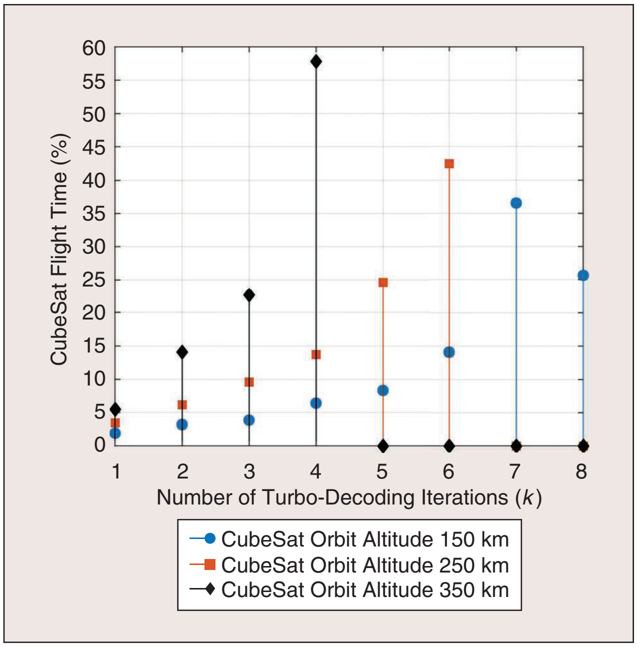

FIGURE 8 The percentage of the CubeSat's flight time versus the number of turbo-decoding iterations allowed in the case of split D of the virtual BBU, with the constraint of at least one turbo-decoding iteration executed onboard the CubeSat.

TABLE 1 ***The CubeSat's flight time for different orbits' altitudes (imposing $k_{min} = 1$).***

CubeSat Orbit Altitude (km)	*Flight Time (s)*
350	128
250	146
150	156

TABLE 2 ***The CubeSat's daily visions and the number of satellites needed to guarantee 24-h service continuity.***

Border Area	*Daily Visions*	*CubeSat Number*
1	5	5
2	6	4
3	7	4

- J is the time required for processing other wireless functions (e.g., synchronization).
- D is the RRH–BBU link delay.
- L is the turbo-coded block length measured in bits.
- F is the number of elementary operations required per decoding iteration.
- T_{proc} is the processing time required by LTE turbo-coded block detection from the RRH (which is given by $L/_{\text{RC-RAN}}$, with $R_{\text{C-RAN}}$ being the bit rate of the dedicated RRH–BBU connection).

In Figure 7, the value of k is plotted according to O and D, assuming $p = 1.2$ GOPS, $\Phi = 3$ ms, $J = 0.9$ m [13], $L = 6{,}144$ bits, (as per LTE standard [13]), $F = 200$ [13], and $R_{\text{C-RAN}} = 181$ Mb/s (the required bit rate for the efficient splitting of L2 high functionalities is 180 Mb/s [6]). It is evident from Figure 7 that the execution of the iterative turbo-decoding operation during the assigned time slot fails when D exceeds 2.1 ms. For lower D, k also increases as O increases. A satisfactory range of values should be k greater and equal to 3, as shown in [13]. Indeed, it can be easily verified that iterative turbo decoding generally converges to very low BERs after a few iterations and for low signal-to-noise ratios when robust turbo codes with low coding rates (one third or one half) are adopted [14]. Following such considerations, the altitude of the CubeSat's orbit, the transmission angle, and the BBU processing architecture should be carefully designed to cope with the PHY performance requirements.

In Figure 8, some preliminary results about the percentage of the CubeSat's flight time versus k are displayed. For this series of results, the transmission angle was computed to satisfy the constraint of having at least a single turbo-decoding iteration during CubeSat's flight time. Moreover, the number of core processors O has been fixed as equal to 6. In Figure 8, we note that, for an orbit altitude of 350 km, the BBU can support k decoding iterations greater and equal to 3 for 80.6% of the CubeSat's flight time. By decreasing the orbit's altitude, the percentage of flight time for which turbo decoding satisfactorily performs increases by up to 90.4 and 94.9% for 250 and 150 km, respectively. The theoretical convenience of lower CubeSat orbits is also confirmed by the results listed in Table 1, relative to the CubeSat's flight time subjected to the constraint of $k_{\min} = 1$; however, decreasing the orbit's altitude is always at the expense of reduced orbit stability and satellite lifetime.

Finally, one may ask how many CubeSats are necessary to guarantee 24-h continuity of service. Table 2 provides an answer to this important question. In this table, the number of daily visions for CubeSats with an altitude lower than 570 km and a sun-synchronous polar orbit was obtained at the latitude of the border areas considered in this article using (1) in [15]. The number of required CubeSat is computed accordingly.

Conclusions

In this article, we studied and described the realization of C-RAN in a system based on UAVs (mobile BSs) and CubeSats when the scope is the realization of reliable wireless networks in complex areas. In particular, we provided the main design guidelines and results necessary to realize split D of virtual BBUs, which were able to perform an FEC (and, consequently, HARQ) in a centralized manner on the satellites.

To the best of our knowledge, the work described in this article is the first that deals with the design of a C-RAN and a virtual BBU split using CubeSats. The importance of this article comes from the growing interest in deploying low-cost satellites and drones to provide connectivity in extremely rural and remote areas. Future work will concern end-to-end performance evaluations in terms of BER and achievable throughput of the system, taking into account the issues and constraints involving the CubeSat's feeder link.

Acknowledgment

The work discussed in this article was partially funded by a North Atlantic Treaty Organization Science for Peace and Security (SPS) Programme, in the framework of project SPS G5428, "Dynamic Architecture based on UAVs Monitoring for Border Security and Safety."

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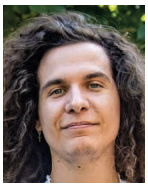

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